From Static to Contextual: Distributional Semantics and What's Inside BERT

NLP CS60075 Assignment II – Project 5, dataset: UD English EWT

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Abstract

We build classical PPMI+SVD word vectors from the UD English EWT training text (Task 1), reproduce the Hewitt & Manning (2019) structural probe on frozen `bert-base-uncased` (Task 2), and turn the most syntactic BERT layer into static word vectors that we compare with Task 1 on semantic and structural tests (Task 3). The probe reproduces the paper almost exactly: 79.5% UUAS at layer 8 on EWT test vs. 79.8% on PTB, with the same mid-layer peak (layers 6–8 are statistically tied). BERT-derived static vectors are far better than PPMI-SVD on meaning (WordSim-353 ρ 0.577 vs. 0.104; analogy 67.7% vs. 0.51%), but *worse* on the tree-recovery check (17.9% vs. 24.8% UUAS on 660 test sentences, close to a random tree). The two evaluations therefore disagree. We show with measured diagnostics why: syntax lives in the context that pooling averages away, and BERT-static similarity is *paradigmatic* (substitutes) while dependency edges need *syntagmatic* association (partners), which the count-based vectors partly have.

1 Introduction

The distributional hypothesis (Harris, 1954) says a word's meaning can be read from the contexts it occurs in. Count-based models turn this into one fixed vector per word (Levy and Goldberg, 2014; Levy et al., 2015); BERT (Devlin et al., 2019) gives every *token* its own vector that depends on the sentence. The project asks three questions on one treebank, UD English EWT (Silveira et al., 2014; de Marneffe et al., 2021): **(T1)** how good are classical PPMI+SVD vectors trained on 200k words, and do they contain any tree structure? **(T2)** can a linear structural probe recover dependency trees from BERT, and which layer ℓ^* encodes them best? **(T3)** if we average layer-ℓ^* vectors into static vectors, are they better than T1 on word meaning, on tree structure, and do the two answers agree?

Main findings. (1) T1 is limited by data: WS-353 $\rho = 0.104$ overall but 0.317 on frequent words; its MST beats a random tree but not the linear chain. (2) The probe reaches 79.5% UUAS / NSpr 0.859 at ℓ^*=8, within 0.3 pt UUAS of the paper; context adds ≈21 pt over the non-contextual layer 0, mostly for function words. (3) T3 wins semantics by a large, significant margin; it loses the structural check; we explain the disagreement and test each explanation quantitatively (§5.1). All numbers in this report are generated by a script from the experiment outputs; seeds are fixed (§8).

2 Data, benchmarks and shared protocol

Treebank. UD English EWT (web text: reviews, e-mails, newsgroups, Q&A, blogs), loaded from the Hugging Face `universal-dependencies/universal_dependencies` repository at the immutable tag `ud2.18-loader-v2.3.0`. We use the syntactic words (multi-word-token surface forms and empty nodes excluded), the `head` field for trees and UPOS for punctuation. All trees are valid; only 10 punctuation tokens in the whole treebank are non-leaves.

split	sentences	words	mean len.	max len.	PUNCT tokens	sents. with MWT
train	12,544	204,578	16.3	159	23,596	2,192
dev	2,001	25,148	12.6	75	3,075	303
test	2,077	25,094	12.1	81	3,096	302

Table 1: UD English EWT splits. Train has 16,654 lowercased word types, 47% of them singletons; 7.5% of test tokens are unseen in train.

Benchmarks. WordSim-353 (353 pairs, Finkelstein et al. 2002) and the Google analogy set (19,544 questions, 14 sections, Mikolov et al. 2013), the copies shipped with `gensim` (pinned commit; SHA-256 recorded), lowercased. **Coverage is the central practical issue**: with a 200k-word corpus only 248/353 WS pairs (70.3%)

and 5,727/19,544 analogy questions (29.3%) have all their words in the vocabulary; the median train frequency of a covered WS word is only 11. T1 and T3 use *the same vocabulary* (all lowercased train types), so their coverage is identical and every comparison is on the same items. We report two numbers for each benchmark: *all items* (the spec-literal number: every one of the 353 pairs, OOV pairs given the median model similarity, a neutral rank; OOV analogy questions counted wrong) and *covered items* (the informative comparison).

Structural check (T1, T3). 20 test sentences, sampled once with seed 42 from the 660 test sentences that have 5–40 non-punctuation words, all in the train vocabulary (a static model has no vector for an OOV word); the same 20 IDs are reused in T3. For each sentence we build Prim's MST over pairwise cosine distances of the words' vectors and score undirected attachment (UUAS) against the gold tree, ignoring punctuation (as in the probe paper). Because 20 sentences (234 edges) are few, we also report **the whole eligible pool of 660 sentences**, bootstrap 95% CIs (1000 resamples of sentences), two baselines – the *linear chain* (attach each word to the next; the paper's LINEAR) and the expected UUAS of a *uniformly random tree* – and paired tests between systems.

3 Task 1 – PPMI + SVD baseline

Method. Word–word co-occurrence counts over the train sentences with a symmetric window of 5 words on each side (within the sentence, unweighted), PPMI = $\max(0, \log \frac{P(w,c)}{P(w)P(c)})$ (Levy and Goldberg, 2014), truncated SVD to 100 dimensions (randomized solver, seed 42), word vectors $U\Sigma$. Similarity = cosine. Analogies by 3CosAdd ($\arg\max_x \cos(x, b - a + c)$ over the vocabulary, the three query words excluded). Beyond the spec we ran the Levy et al. (2015) hyper-parameters (context-distribution smoothing $\alpha = 0.75$, eigenvalue weight $p \in \{0, 0.5\}$) and all-but-the-top post-processing, to make sure T3 is compared with a *fair* baseline.

	WordSim-353 ρ			Analogy acc. (%)		MST UUAS (%)	
PPMI-SVD variant	covered (248) [95% CI]	all 353	freq. ≥20	covered	all	20 sent.	pool (660)
spec: α=1, p=1 (UΣ)	0.104 [−0.03, 0.22]	0.048	0.317	0.51	0.15	20.5	24.8
α=0.75, p=1	0.060 [−0.07, 0.20]	0.043	0.199	0.65	0.19	15.8	20.8
α=1, p=0.5	0.129 [−0.00, 0.24]	0.091	0.289	0.37	0.11	20.9	24.6
α=1, p=0 (U)	0.134 [0.00, 0.25]	0.089	0.320	0.21	0.06	22.6	24.5
α=0.75, p=0.5	0.097 [−0.04, 0.22]	0.066	0.174	0.42	0.12	16.2	20.8
spec + all-but-the-top (D=1)	0.153	0.103	0.294	0.42	–	20.5	24.5
linear chain (attach to next word)	–	–	–	–	–	43.2	43.0
random tree (expected)	–	–	–	–	–	15.2	17.1

Table 2: Task 1 results. "covered" = items whose words are all in the vocabulary; "all" = all items (spec-literal). "freq. ≥20" = the 32 covered pairs whose two words occur ≥20 times in train. Pool = all 660 eligible test sentences.

Results (Table 2). WS-353 $\rho = 0.048$ over all 353 pairs and 0.104 [−0.03, 0.22] on the covered ones; analogy accuracy 0.15% (all) / 0.51% (covered; 29 of 5,727 correct). The structural check gives UUAS 20.5% [16.3, 26.2] on the 20 sentences: +5.3 pt above a random tree (paired CI [1.4, 9.2], significant) but −22.6 pt below the linear chain (43.2%).

Discussion. The numbers are low, but this is a data effect, not an implementation error. Evidence: (i) restricted to frequent words the same vectors are reasonable – ρ rises from 0.104 (all covered pairs) to 0.248 (both words ≥5 occurrences), 0.283 (≥10) and 0.317 (≥20); (ii) the nearest neighbours of frequent words are sensible co-occurrence partners (Table 9); (iii) no hyper-parameter rescues it: the best variant ($p = 0$) reaches $\rho = 0.134$ and no setting gets the analogies above 1%. Analogies need precise linear offsets between *four* reliably-estimated vectors, which 200k words cannot give (the in-vocabulary words of the capital, family, comparative and currency sections have median train counts of 6–11). On trees, PPMI-SVD recovers relations between content words that co-occur (amod 0.40, advmod 0.36, conj 0.34, obj 0.31 recall on the pool) but almost no function-word attachments (cc 0.05, case 0.12, mark 0.13, det 0.14): function words co-occur with everything, so their vectors are uninformative.

4 Task 2 – Reproducing the structural probe

Features. Each EWT sentence is fed to `google-bert/bert-base-uncased` (revision `86b5e09`) as pre-split words; all 13 hidden states are kept (layer 0 = embedding layer, layers 1–12 = transformer blocks, i.e. HF `hidden_states[`ℓ`]`); a word's vector is the mean of its word-piece vectors, as in the paper. No sentence

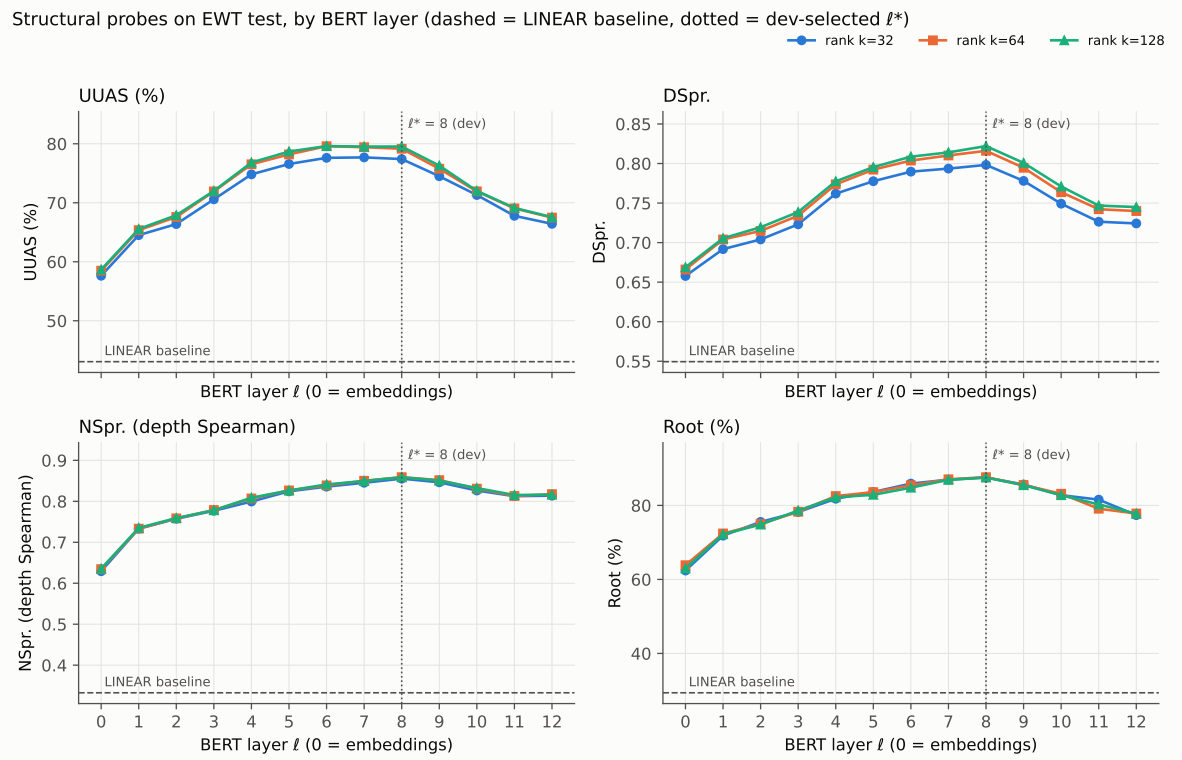


Figure 1: Probe metrics on EWT test vs. BERT layer, one line per rank (dashed = LINEAR baseline, dotted = dev-selected ℓ^*).

exceeds 510 word pieces (max. 335), so no truncation is needed. Features are cached once (float16, 5.1 GB, 410 s on one L40S).

Probes. Distance probe $d_B(h_i, h_j) = \|B(h_i - h_j)\|_2^2$ trained to match the gold tree distance; depth probe $\|Bh_i\|_2^2$ trained to match the gold depth; $B \in \mathbb{R}^{k \times 768}$, $k \in \{32, 64, 128\}$, $B \sim \mathcal{U}(-0.05, 0.05)$. Loss: L1 between prediction and gold, summed over word pairs (words) and divided by n^2 (n) per sentence, averaged over the batch – the paper's loss. Gold distances/depths come from `head` by BFS over all tokens. Training: Adam, lr 10^{-3}, batches of 20 sentences, 30 epochs, train split, 13 layers × 3 ranks × 2 probes = 78 probes (layer 0 added as a non-contextual baseline). As in the paper (App. A), when the dev loss does not improve, the learning rate is multiplied by 0.1 (optimizer reset), and the best-dev checkpoint is kept; see §4.2 for why this matters. **Evaluation** follows the official `reporter.py`: UUAS = MST of predicted distances vs. the gold tree over non-punctuation words, micro-averaged over all test sentences; DSpr = per-word Spearman of predicted vs. gold distance rows; NSpr ("depth Spearman") = per-sentence Spearman of predicted vs. gold depth; both averaged per sentence length and macro-averaged over lengths 5–50; Root% = the shallowest non-punctuation word is the root. ℓ^* is chosen on **dev** UUAS (best rank per layer); test is never used for a choice.

Results (Fig. 1, Table 3). UUAS rises from 58.7% at layer 0 to a plateau over layers 6–8 and falls to 67.6% at layer 12; the depth probe has the same shape. Dev selects ℓ^*=8 (dev UUAS 80.0%; depth also peaks at 8 on dev and test). On test, layers 6–8 are statistically indistinguishable from 8 (test argmax is layer 6 by 0.08 pt), every other layer is significantly worse. **Rank**: at ℓ^*, UUAS 77.4 / 79.2 / 79.5% for $k = 32/64/128$ and NSpr 0.855 / 0.859 / 0.859 – the gain stops around $k = 64$, as in the paper (their Fig. 5): the tree lives in a small (≈64-dim.) subspace of the 768 dimensions. Every probe reached its best dev loss by epoch 8–17.

What context adds. Layer 0 (word-piece + position embeddings, no context) already reaches 58.7%, far above LINEAR (43.1%) and close to the paper's non-contextual PROJ0 (59.8): word identity plus position explains a large part of a tree. The remaining ≈21 pt are the evidence that BERT's context encodes syntax. The gains are largest for function words, which have no lexical cue to their head (recall layer 0 → 8: case 0.35→0.73, cc 0.40→0.82, mark 0.56→0.80, obl 0.42→0.69), and smallest for adjacent pairs (compound 0.82→0.87, amod 0.76→0.87). At ℓ^* the easiest relations are obj (0.93), det (0.91), aux (0.89); the hardest are conj (0.56) and obl (0.69) – UD attaches every conjunct to the first one (long arcs) and obl is the PP-attachment ambiguity (per-relation heatmap in App. B).

	UUAS (%), rank k				ΔUUAS vs. ℓ^* (pt)	NSpr (depth Spearman), k			
ℓ	32	64	128	DSpr	paired 95% CI (k=128)	32	64	128	Root%
0	57.6	58.5	58.7	0.669	−20.9 [−21.6, −20.1]	0.629	0.634	0.636	63.0
1	64.5	65.4	65.6	0.705	−14.0 [−14.7, −13.2]	0.733	0.733	0.735	72.2
2	66.4	67.6	67.9	0.719	−11.6 [−12.4, −10.9]	0.757	0.758	0.759	74.8
3	70.6	71.9	72.0	0.739	−7.5 [−8.2, −6.8]	0.777	0.779	0.778	78.3
4	74.8	76.5	76.8	0.777	−2.7 [−3.3, −2.1]	0.799	0.807	0.809	82.2
5	76.6	78.2	78.7	0.795	−0.8 [−1.4, −0.3]	0.824	0.827	0.826	82.9
6	77.6	79.6	79.6	0.809	+0.1 [−0.4, 0.6]†	0.836	0.839	0.841	84.8
7	77.7	79.4	79.5	0.814	0.0 [−0.4, 0.5]†	0.845	0.850	0.850	87.0
8	77.4	79.2	79.5	0.822	–	0.855	0.859	0.859	87.5
9	74.5	75.8	76.3	0.801	−3.2 [−3.7, −2.8]	0.846	0.852	0.851	85.5
10	71.3	71.9	72.0	0.771	−7.5 [−8.1, −7.0]	0.826	0.831	0.832	82.8
11	67.8	69.1	69.1	0.742	−10.4 [−11.0, −9.8]	0.812	0.813	0.815	80.3
12	66.4	67.5	67.6	0.745	−12.0 [−12.5, −11.4]	0.814	0.818	0.816	77.8
LINEAR		43.1		0.549	–		0.333		29.4

Table 3: Task 2 on EWT test, all layers and ranks. DSpr/Root% use the dev-best rank of the layer. The Δ column is a paired bootstrap of each layer vs. ℓ^* over the same sentences; † = 95% CI contains 0 (not significantly different). Shaded: ℓ^*.

4.1 Comparison with the paper

	system	UUAS	DSpr	Root%	NSpr
Paper (PTB WSJ)	LINEAR	48.9	0.58	2.9	0.27
Paper (PTB WSJ)	PROJ0	59.8	0.73	64.4	0.75
Paper (PTB WSJ)	ELMO1	77.0	0.83	86.5	0.87
Paper (PTB WSJ)	BERTBASE7	79.8	0.85	88.0	0.87
Paper (PTB WSJ)	BERTLARGE16	81.7	0.87	90.1	0.89
Ours (EWT)	LINEAR (word order only)	43.1	0.549	29.4	0.333
Ours (EWT)	BERT layer 0 (embeddings + position)	58.7	0.669	63.0	0.636
Ours (EWT)	**BERT-base-uncased layer 8** (paper schedule)	79.5	0.822	87.5	0.859
Ours (EWT)	BERT layer 8, spec-literal (constant lr, last epoch)	62.0	0.760	58.6	0.803

Table 4: Our numbers vs. Hewitt and Manning (2019) Table 1 (PTB WSJ test, BERT-base *cased*, Stanford Dependencies).

Our best probe is within 0.3 pt UUAS, 0.03 DSpr, 0.5 pt Root% and 0.01 NSpr of the paper's BERTBASE7 – the result reproduces. Our layer-0 probe matches their non-contextual PROJ0 baseline (UUAS 58.7 vs. 59.8). The paper numbers BERT-base layers 0–11 without the embedding layer, so its "layer 7" is our HF layer 8 (or 7 under the other reading); since layers 6–8 are tied, the comparison holds either way. Likely causes of the small DSpr/NSpr gap, none of which we can isolate without more runs: (1) **data scale**: 12.5k train sentences vs. ≈40k in PTB; (2) **genre**: web text with typos and fragments vs. edited news; (3) **annotation**: UD's content-head trees (prepositions and copulas are dependents) vs. Stanford Dependencies, which changes the gold distance geometry; (4) **uncased** vs. cased model; (5) **budget**: rank ≤128 and 30 epochs vs. full rank and up to 40 epochs. The LINEAR baseline differs more (UUAS 43.1 vs. 48.9; Root% 29.4 vs. 2.9): EWT sentences are short and often *start with their root* (imperatives "Say . . . ", "Thanks . . . ", fragments) whereas WSJ sentences start with a subject.

4.2 Control run: the assignment's literal training schedule

The assignment specifies Adam with lr 10^{-3} for 30 epochs; we added the paper's learning-rate decay and best-dev selection. To make sure this deviation does not drive our conclusions we retrained every layer ($k = 128$) with a constant lr 10^{-3} and kept the epoch-30 probe. **ℓ^* is unchanged (8 on dev and test) and the curve keeps its rise–peak–fall shape**, but UUAS at ℓ^* drops to 62.0% (DSpr 0.760, NSpr 0.803, Root 58.6%; Table 4). The constant-lr probe does not converge: at layer 8 its *training* loss stays at 0.62 (vs. 0.42 with decay) and its dev loss oscillates between 0.54 and 0.59 for all 30 epochs. With an L1 loss the gradient does not shrink near the optimum, so Adam's steps stay of size ≈lr and circle the minimum; the decay is what lets the probe converge. The deviation is therefore necessary for a faithful reproduction, and the main conclusion (ℓ^*=8) does not depend on it.

system	WordSim-353 ρ covered [95% CI]	WordSim-353 ρ all 353	Analogy acc. (%) covered	Analogy acc. (%) sem / syn	Analogy acc. (%) all	MST UUAS (%) 20 sent. [95% CI]	MST UUAS (%) pool
T1 PPMI-SVD (spec)	0.104 [−0.03, 0.22]	0.048	0.5	2.1 / 0.3	0.1	20.5 [16.3, 26.2]	24.8
T1 PPMI-SVD, best variant (p=0)	0.134 [0.00, 0.25]	0.089	0.2	0.9 / 0.1	0.1	22.6 [17.8, 28.4]	24.5
T3 BERT-static, layer 8	0.577 [0.47, 0.66]	0.452	67.7	74.5 / 66.6	19.8	18.4 [13.6, 26.0]	17.9
T3 + all-but-the-top (D=7)	0.682 [0.61, 0.74]	0.539	71.0	79.0 / 69.7	20.8	19.7 [14.9, 26.2]	19.6
T2 probe, layer 8 (contextual, supervised; reference)	–	–	–	–	–	81.2 [75.9, 85.9]	82.9
linear chain / random tree	–	–	–	–	–	43.2 / 15.2	43.0 / 17.1

Table 5: Task 3 vs. Task 1 (identical items), with the Task 2 probe as a reference on the same 20 sentences and pool.

A − B	WordSim-353 $\Delta\rho$ [95% CI]	Analogy (McNemar) only A / only B	Analogy (McNemar) p	MST ΔUUAS (pt) [95% CI] 20 sentences	MST ΔUUAS (pt) [95% CI] pool (660)
T3 − T1	+0.474 [0.36, 0.61]	3,848 / 0	$< 10^{-300}$	−2.1 [−6.4, 3.2]	−6.9 [−8.0, −5.8]
T3+ABTT − T1	+0.579 [0.46, 0.71]	4,036 / 0	$< 10^{-300}$	−0.9 [−6.6, 5.6]	−5.1 [−6.2, −4.1]
T3+ABTT − T3	+0.105 [0.05, 0.16]	276 / 88	1×10^{-23}	+1.3 [−3.2, 5.6]	+1.7 [1.0, 2.6]
T2 probe − T3	–	–	–	+62.8 [53.2, 69.6]	+65.0 [63.8, 66.3]

Table 6: Paired tests on identical items. Analogy: number of questions solved by only A / only B, exact McNemar p.

5 Task 3 – BERT-derived static vectors

Method. For each of the 16,654 T1 word types we average the layer-ℓ^* vectors (each the mean of its word pieces, computed *in context*) over all its occurrences in EWT train – the mean/mean pooling that Bommasani et al. (2020) found best. The vocabulary and row order equal T1's, and WS-353, analogy and the 20-sentence MST are run by *the same function* as T1. We add: (a) **all-but-the-top** (Mu and Viswanath, 2018): subtract the mean vector and remove the top $D = d/100$ principal directions ($D = 7$ for BERT; also applied to T1 with $D = 1$ so that post-processing is not credited to BERT), because pooled BERT vectors are anisotropic (Ethayarajh, 2019); (b) **paired significance tests** on identical items (bootstrap of $\Delta\rho$ over pairs, exact McNemar test over analogy questions, bootstrap of ΔUUAS over sentences); (c) a **layer sweep** 0–12; (d) a **diagnostic ladder** of MSTs on the same sentences; (e) two combination variants (concatenated L2-normalised [T1; T3] vectors, and T3 vectors seen through the T2 probe matrix B).

Word meaning: T3 is much better (Tables 5, 6). WS-353 ρ rises from 0.104 to 0.577 on covered pairs ($\Delta\rho$ = +0.474, CI [0.36, 0.61]) and from 0.048 to 0.452 on all 353; all-but-the-top adds another +0.105 (0.682). Analogy accuracy goes from 0.51% to 67.7% (71.0% with ABTT) on covered questions and to 19.8% on all 19,544: T3 solves 3,848 questions T1 misses and T1 solves 0 that T3 misses. Unlike T1, T3 is good even on rare words (ρ = 0.577 on all covered pairs vs. 0.661 on frequent ones), because BERT's knowledge comes from pre-training, not from the 200k EWT words – the comparison is "counting a small corpus" vs. "a model pre-trained on ≈3.3B words", so this win is expected. Our numbers agree with the literature: our best layer for WS-353 is layer 1 with ρ = 0.690 from ≈6.8k contexts; Bommasani et al. (2020) report 0.683 at layer 1 for bert-base-uncased with 10k contexts.

Tree structure: T3 is worse. On the 20 required sentences T3 gets 18.4% vs. T1's 20.5% (Δ =-2.1 pt, CI [−6.4, 3.2]: *not significant* – 20 sentences cannot separate them). On the 660-sentence pool the difference is significant: 17.9% vs. 24.8% (Δ =-6.9 pt, CI [−8.0, −5.8]), barely above a random tree (17.1%). The probe on the same sentences reaches 81.2% / 82.9%. **So the two evaluations disagree**: pooling from the syntax layer gives much better vectors by the standard benchmarks and worse ones by the structural check.

Is the syntax layer the right layer for static semantics? No (Fig. 2). Static similarity is best at the lowest layers: WS-353 peaks at layer 1 (0.690) and analogies at layer 2 (80.3%); even layer 0, essentially BERT's input embedding table with positions averaged out, gives ρ = 0.685 and 72.1%. This matches Bommasani et al. (2020) (first quarter of the layers is best). Raw WS falls with depth while anisotropy (mean cosine of random word pairs) rises from 0.04 to 0.40 (layer 11); after all-but-the-top the WS curve is almost flat (0.64–0.71), so much of the decline is the "narrow cone" effect, not lost meaning. Analogies still decline after ABTT: upper layers are more context-specific. No layer makes static vectors good at trees (15–21% everywhere).

5.1 Why the evaluations disagree – measured, not assumed

1. The tree lives in the context, and pooling averages it away. The ladder (Table 7) adds one ingredient per rung: static T3 17.9% → the same static vectors through the probe's syntactic subspace B 35.9% → per-token

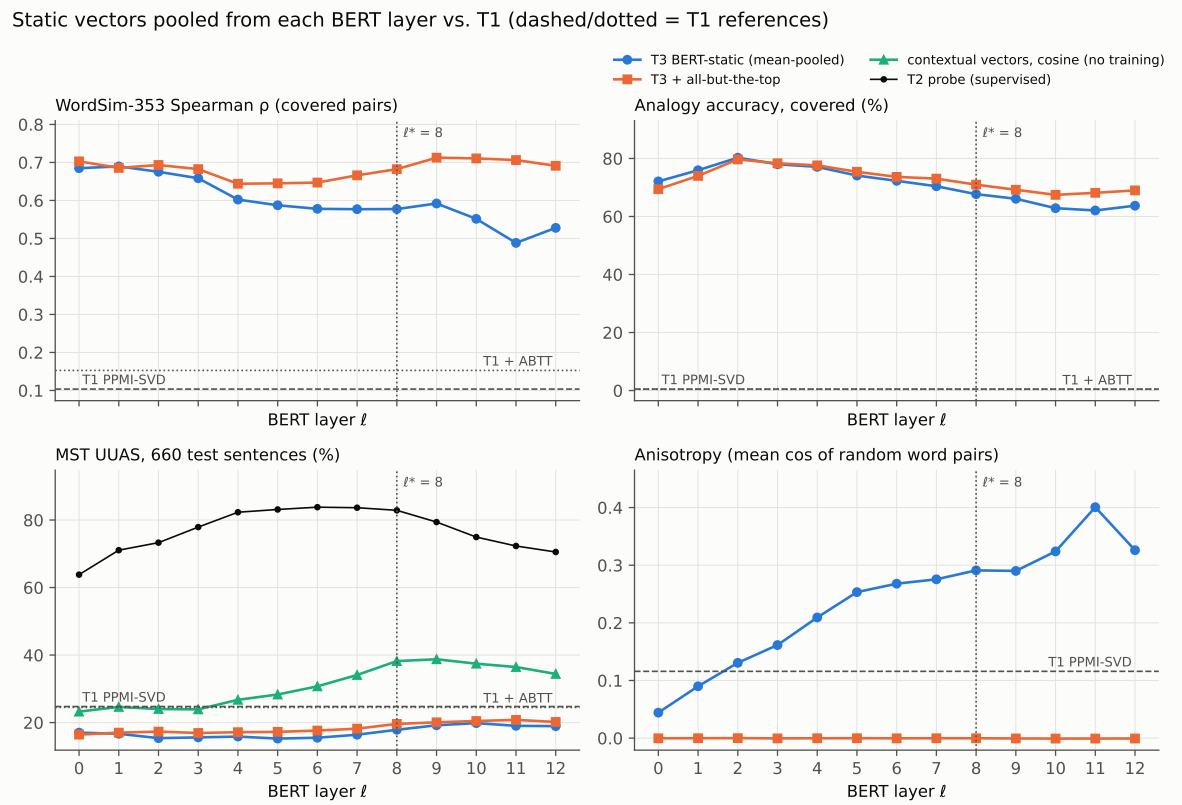


Figure 2: Static vectors pooled from every BERT layer (raw and with all-but-the-top) vs. T1 references; the MST panel also shows raw contextual vectors (cosine, no training) and the T2 probe on the same 660 sentences. Numbers in App. A.

contextual vectors with plain cosine, no training, 38.2% → the trained probe 82.9%. Averaging "bank" over all its sentences gives a good dictionary entry, but it cannot know what "bank" attaches to in *this* sentence. Even the raw contextual vectors stay below the linear chain (43.0%): BERT's tree is not in its raw cosine geometry but in a learned low-rank subspace – exactly what the structural probe claims.

2. "Similar" is not "connected" (paradigmatic vs. syntagmatic). T3's 10 nearest neighbours share the word's majority part of speech 0.74 of the time (T1: 0.33; random pairs: 0.22) – they are *substitutes* (good → excellent, great; Table 9). T1's neighbours actually co-occur with the word within ±5 words 0.56 of the time (T3: 0.16; random: 0.07) – they are *partners* (said → spokesman; car → driving). This is the classic paradigmatic/syntagmatic distinction (Sahlgren, 2006). The direct test is the edge AUC: **inside a sentence, T3 closeness carries no information about which words are connected (AUC 0.46, slightly below chance), T1 a little (0.54)**; contextual vectors (0.70), word order alone (0.75) and the probe (0.96) are informative. Per relation, T1 beats T3 on head–dependent pairs such as nsubj (0.23 vs. 0.09), obj (0.31 vs. 0.15), amod (0.40 vs. 0.16); T3 wins only on conj (0.44 vs. 0.34), the one relation that links two *similar* words ("cats and dogs"). A first guess of ours – "gold edges join different categories, so same-category similarity is the wrong signal" – is *not* supported: gold edges are same-UPOS slightly *more* often than non-edges (0.13 vs. 0.09); what matters is the AUC result above.

3. Within-sentence hubs. Over the whole vocabulary T3 is *not* more hub-prone than T1 (skewness of the 10-NN in-degree N_{10} (Radovanović et al., 2010): 1.45 vs. 2.10). Inside sentences, however, T3's MSTs are star-shaped around frequent function words: function-word tokens get 2.29 edges on average (gold 1.05, T1 1.84) and the largest degree per sentence is 5.46 (gold 4.53). All-but-the-top reduces both (Table 8) and gives the significant +1.7 pt on the pool, but does not fix the tree.

Combination variants. Concatenating [T1; T3] hurts semantics (WS 0.397, analogy 22.2% – T1's noisy similarities dilute BERT's) and only matches T1 on trees (25.1%): the two spaces are not usefully complementary. Projecting T3 through the probe's B doubles its tree score (35.9%): the syntactic subspace carries some *type-level* information (which kind of word attaches to which), but most of the tree requires context.

representation for the MST	tier	20 sentences [95% CI]	pool (660) [95% CI]	edge AUC
random tree (expected)	–	15.2	–	–
linear chain	–	43.2 [40.6, 46.4]	43.0 [42.4, 43.7]	0.75
T1 PPMI-SVD (spec)	req.	20.5 [16.3, 26.2]	24.8 [23.7, 26.0]	0.54
T1 PPMI-SVD (p0)	B	22.6 [17.8, 28.4]	24.5 [23.4, 25.6]	–
T3 BERT-static L8	req.	18.4 [13.6, 26.0]	17.9 [16.8, 18.9]	0.46
T3 BERT-static L8 + ABTT	B	19.7 [14.9, 26.2]	19.6 [18.6, 20.7]	0.44
concat [T1 ; T3] (L2-normalised)	C	19.2 [15.2, 25.1]	25.1 [24.0, 26.2]	–
T3 BERT-static L8 through probe B	C	29.5 [23.7, 38.2]	35.9 [34.3, 37.5]	–
contextual L8, cosine (no training)	B	35.9 [32.1, 39.6]	38.2 [37.3, 39.2]	0.70
T2 distance probe L8 k=128	ref.	81.2 [75.9, 85.9]	82.9 [81.9, 83.9]	0.96

Table 7: The diagnostic ladder on the same sentences. Edge AUC: within each pool sentence, the probability that a gold edge pair is closer than a non-edge pair (0.5 = no information), averaged over sentences. Tier: req. = required, B/C = our extensions.

	T1 PPMI-SVD	T3 layer 8	T3 + ABTT	random / gold
NN with same majority UPOS (top-10)	0.326	0.740	0.710	0.223
NN that co-occur within ±5 in train	0.557	0.162	0.156	0.067
hubness: skewness of N_{10}	2.10	1.45	0.57	–
function words among top-50 hubs	0.10	0.12	0.02	–
edge AUC (gold edge closer than non-edge)	0.54	0.46	0.44	0.50
MST: mean degree of function-word tokens	1.84	2.29	2.04	1.05 (gold)
MST: mean max. degree per sentence	3.75	5.46	3.62	4.53 (gold)
MST: edges joining same-UPOS words	0.21	0.25	0.29	0.13 (gold)

Table 8: Diagnostics (`diagnostics_task3.py`). NN statistics use the 2,134 types with train frequency ≥10 and the majority UPOS of each type; hubness uses the 3,919 types with frequency ≥5; MST statistics use the 660-sentence pool.

6 Qualitative analysis

Task 2. Examples are chosen by rank, not by hand: among test sentences with 10–18 words, the best, the median and the worst UUAS (Fig. 3 shows the last two). *Good* (UUAS 1.00): "I have eaten here several times and everytime the service is slower than slow." – every edge correct. *Typical* (0.82): "Say after I finished those 2 years and I found a job."; missed: Say-finished (discourse), finished-found (conj); spurious: Say-after, I-I. The probe joins the two subjects "I" to each other instead of linking the clauses through their verbs, and misreads the discourse use of "Say". *Poor* (0.30): the verbless itinerary "Raleigh to Chicago, Chicago to Seoul and then Seoul to Manila."; the probe links *repeated identical words* (to–to, Chicago–Chicago) and chains the cities left to right, whereas UD attaches each conjunct to the first. Identical word forms get very similar BERT vectors, so lexical identity leaks into the syntactic distance – the most frequent failure we saw, together with coordination and PP attachment.

Task 3. Neighbours (Table 9) show the paradigmatic/syntagmatic contrast directly; T1's lists for mid-frequency words (*two*, *computer*) are noise from rare co-occurrences. Analogies (Table 10): T3 is near-perfect on regular morphology (plural 96.6%) and frequent capitals (93.6%), but fails where the answer is ambiguous or rare: currency (0/28; "real" occurs 53 times in EWT train, always as an adjective) and opposites (26.9%). Its failures are usually *near misses* that pick a semantic neighbour over the morphological target (cooler → nicer, biggest → largest, english → british) – again the substitute bias. T1's wrong answers are unrelated co-occurring words (Baghdad:Iraq::Beijing → *jintao*). Figure 4 shows one *representative* sentence of the 20 (chosen as the sentence whose per-method scores are closest to each method's average): T1 and T3 each get 25%, raw contextual vectors 33%, the probe 83%; the T3 tree is a star around "a" and "of".

7 Discussion

What worked. The structural probe reproduces on a different treebank, genre and annotation scheme: same peak layers, same rank saturation, within 0.3 pt UUAS. Pooled BERT vectors are strong lexical-semantic vectors even from 200k words of text, because the knowledge was learned in pre-training; one cheap post-processing step (all-but-the-top) adds a significant +0.1 ρ. Paired tests on identical items, a 660-sentence pool and bootstrap CIs made it possible to separate real effects from noise (e.g. the 6–8 layer plateau; the non-significant 20-sentence T1/T3 difference).

What did not work, and why. (1) PPMI-SVD on 200k words cannot do analogies and only weakly captures

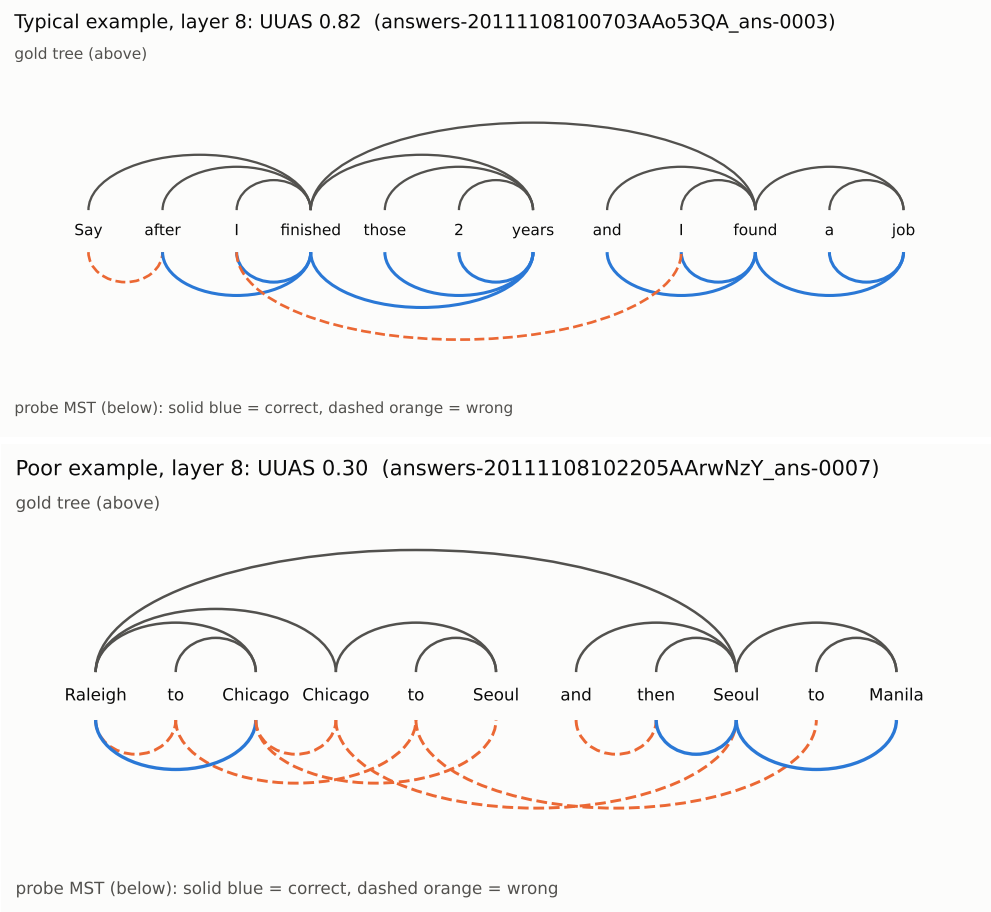


Figure 3: T2 probe (layer 8) on the median (top) and the worst (bottom) test sentence of 10–18 words. Gold tree above the words, probe MST below: blue solid = correct, orange dashed = wrong.

similarity; the cause is data sparsity, confirmed by the frequency-stratified analysis and by the failure of all Levy et al. variants. (2) The task's hypothesis – that pooling from the *syntax-encoding* layer yields better vectors – holds for semantics but not for the structural check. The structural check rewards *syntagmatic* association between words of one sentence; averaging over contexts removes the token-specific information that makes BERT syntactic, and what remains is substitute-style similarity that is uninformative about attachment (edge AUC 0.46). The choice of layer is also mismatched: the syntax layer (8) is not the best static-semantics layer (1–2). (3) The spec's constant learning rate does not let an L1-loss probe converge.

Do the two answers agree? No. "Good word vectors" and "good syntax" are different properties: semantic benchmarks measure paradigmatic similarity between word *types*, the tree check measures syntagmatic relations between *tokens*. A representation can – and here does – improve on one while getting worse on the other.

Limitations. One treebank and one model; the 20-sentence check is under-powered (hence the pool); the MST sample is biased towards in-vocabulary, 5–40-word sentences; BERT brings 3.3B words of external knowledge, so T1 vs. T3 is not a same-data comparison; DSpr/NSpr gap causes are plausible but not isolated.

8 Reproducibility, seeds and deviations

Seeds. `SEED = 42` (`code/common/config.py`) seeds Python, NumPy and PyTorch (`PYTHONHASHSEED` set); the 20-sentence sample (`numpy.random.default_rng(42)`), the random-tree baseline, all bootstraps (1000 resamples), the SVD solver (`random_state=42`) and the anisotropy sampling use it; probe (ℓ, k, type) is initialised with seed $42 + 1000\ell + k$ (+500 for depth) and batch order with seed $42 + \ell$. **Environment.** Python 3.12, PyTorch 2.15, Transformers 5.22, Datasets 5.1 (all versions pinned in `requirements.txt`); 2×NVIDIA L40S (shared). Runtime: T1 seconds; BERT features 7 min; 78 probes 49+54 min on two GPUs; T3 <5 min. **Pretrained checkpoint:** `google-bert/bert-base-uncased`, Hugging Face Hub, revision `86b5e09`. Code: `code/task1`, `task2`, `task3` (plus `common/` for data loading and the shared evaluation); a Colab notebook and `run_all.py` run everything end-to-end (README).

word	T1 PPMI-SVD: 5 nearest neighbours	T3 BERT-static (layer 8): 5 nearest neighbours
good	nice, food, pretty, great, really	excellent, great, nice, decent, wonderful
bad	faith, big, too, wrong, something	terrible, horrible, poor, good, shitty
monday	friday, 0832, july, sunday, 6	tuesday, sunday, thursday, wednesday, saturday
dog	got, horse, cat, myself, leave	dogs, cat, puppy, animal, cats
car	home, guy, test, driving, drive	cars, vehicle, vehicles, suv, ferrari
money	save, enough, much, else, attention	cash, funds, dollars, cashout, bucks
said	spokesman, nun, penman, gregg, maviglio	says, stated, saying, told, say
happy	definitely, customer, guys, !!, safe	pleased, glad, unhappy, satisfied, delighted
computer	disc, plans, evidence, read, mohammed	laptop, computers, pc, machine, internet
two	screamed, paddled, expressway, kitchens, revolt	three, four, 2, five, six

Table 9: Nearest neighbours (cosine). T1 finds co-occurring partners (and noise from rare words); T3 finds substitutes.

section	question (a : b :: c : **gold**)	T3 answer	T1 answer
capital-common-countries	baghdad : iraq :: beijing : **china**	china ✓	jintao
capital-world	amman : jordan :: baghdad : **iraq**	iraq ✓	threats
city-in-state	houston : texas :: philadelphia : **pennsylvania**	pennsylvania ✓	franchise
family	boy : girl :: brother : **sister**	sister ✓	cousin
gram1-adjective-to-adverb	amazing : amazingly :: complete : **completely**	completely ✓	digest
gram2-opposite	acceptable : unacceptable :: aware : **unaware**	unaware ✓	socially
capital-world	beijing : china :: kingston : **jamaica**	jamaican	northwest
currency	brazil : real :: canada : **dollar**	actual	church
gram2-opposite	acceptable : unacceptable :: certain : **uncertain**	specific	socially
gram3-comparative	bad : worse :: cool : **cooler**	nicer	dried
gram4-superlative	bad : worst :: big : **biggest**	largest	chinese
gram6-nationality-adjective	australia : australian :: england : **english**	british	soper

Table 10: Analogy examples: top, questions T3 solves and T1 does not (one per section); bottom, T3 failures.

Deviation / interpretation	Justification
Window "size 5" = 5 words each side	Standard reading (Levy et al., 2015); counts never cross sentence boundaries.
Lowercasing (T1, T3, benchmarks)	The model is uncased; T1 and T3 then share one vocabulary, so their coverage is identical.
WS-353 reported on all 353 pairs *and* on the 248 covered	"All 353" is the spec-literal number (OOV pairs get the median similarity, a neutral rank; a 0 would be non-neutral for anisotropic BERT); "covered" is the informative comparison. Analogy: OOV questions count wrong in "all".
20-sentence sample: 5–40 words, all in vocabulary; punctuation ignored	A static model has no vector for an OOV word; punctuation is excluded exactly as in the probe paper. The bias towards simpler sentences is acknowledged.
LR ×0.1 on dev plateau + best-dev checkpoint (T2)	The paper's schedule; the spec-literal control (§4.2) keeps ℓ^* but does not converge.
Layer 0 probed in addition to 1–12	Non-contextual baseline (cf. the paper's PROJ0).
ℓ^* chosen on dev, not test	Avoids test-set selection; the 6–8 plateau makes the choice robust.
Extra analyses (pool, CIs, paired tests, sweep, ABTT, ladder, diagnostics)	Additions only; every required number is reported as specified.

Table 11: Deviations from the task specification and their justification.

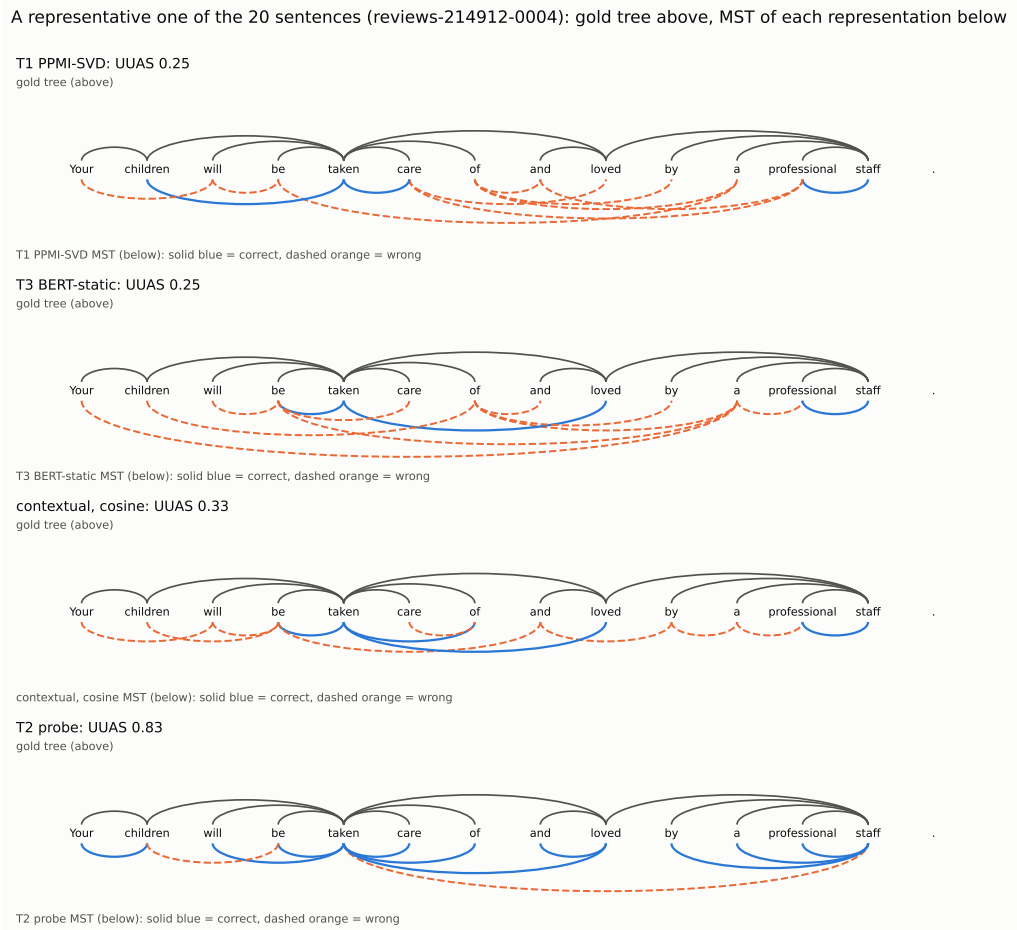


Figure 4: "Your children will be taken care of and loved by a professional staff." – gold tree (above) and the MST of each representation (below).

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A Additional tables

	WS-353 ρ (cov.)		Analogy (cov., %)		MST UUAS on the pool (%)				
ℓ	raw	ABTT	raw	ABTT	static raw	static ABTT	contextual cos.	T2 probe	anisotropy
0	0.685	0.703	72.1	69.4	17.0	16.4	23.2	63.8	0.04
1	0.690	0.686	75.9	74.0	16.7	17.0	24.6	71.1	0.09
2	0.675	0.693	80.3	79.7	15.4	17.3	24.0	73.3	0.13
3	0.658	0.682	78.0	78.3	15.6	16.9	23.9	77.9	0.16
4	0.602	0.644	77.1	77.6	15.9	17.2	26.7	82.3	0.21
5	0.587	0.645	74.1	75.4	15.3	17.3	28.3	83.1	0.25
6	0.578	0.647	72.3	73.7	15.5	17.6	30.7	83.8	0.27
7	0.577	0.666	70.4	73.0	16.4	18.2	34.1	83.6	0.28
8	0.577	0.682	67.7	71.0	17.9	19.6	38.2	82.9	0.29
9	0.592	0.713	66.1	69.2	19.2	20.1	38.8	79.4	0.29
10	0.552	0.711	62.9	67.5	19.8	20.5	37.4	75.0	0.32
11	0.488	0.706	62.1	68.2	19.0	20.8	36.5	72.3	0.40
12	0.528	0.691	63.7	69.0	18.9	20.2	34.4	70.5	0.33
T1	0.104	0.153	0.5	0.4	24.8	24.5	–	–	0.12

Table 12: Layer sweep for pooled static vectors (covered items; pool = 660 sentences), with contextual references.

B Additional figures

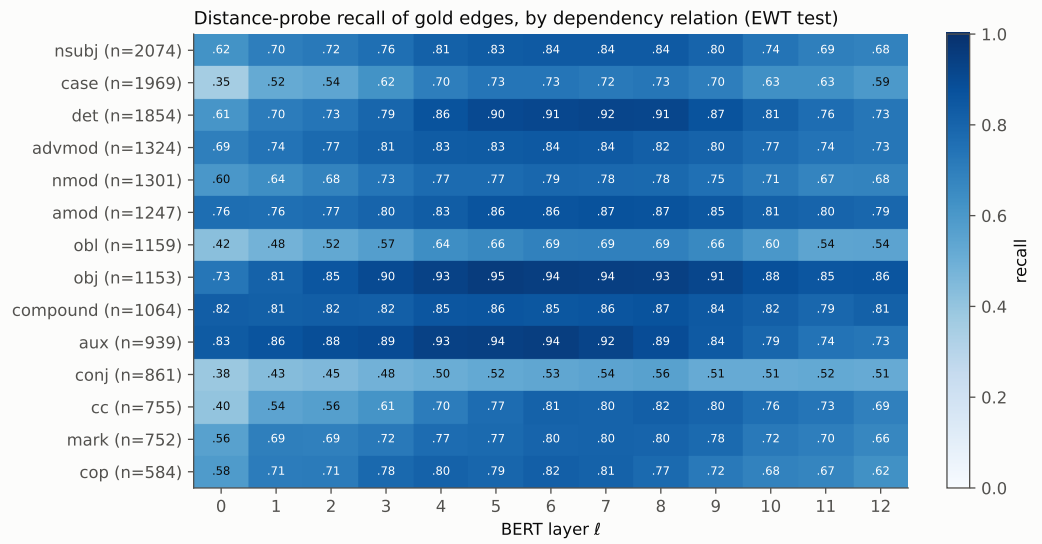


Figure 5: T2 distance-probe recall per dependency relation and BERT layer (EWT test, dev-best rank per layer).

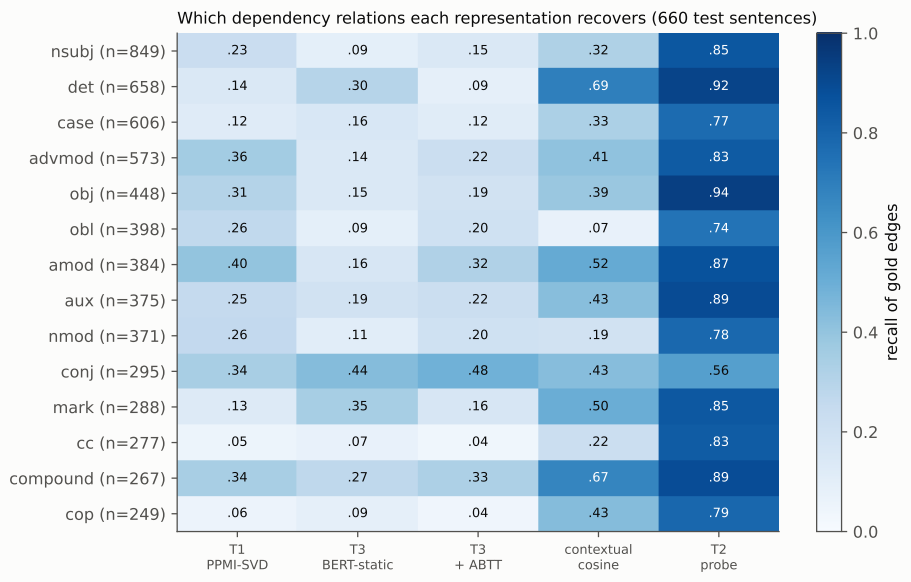


Figure 6: Recall of gold edges per relation for each representation on the 660-sentence pool.

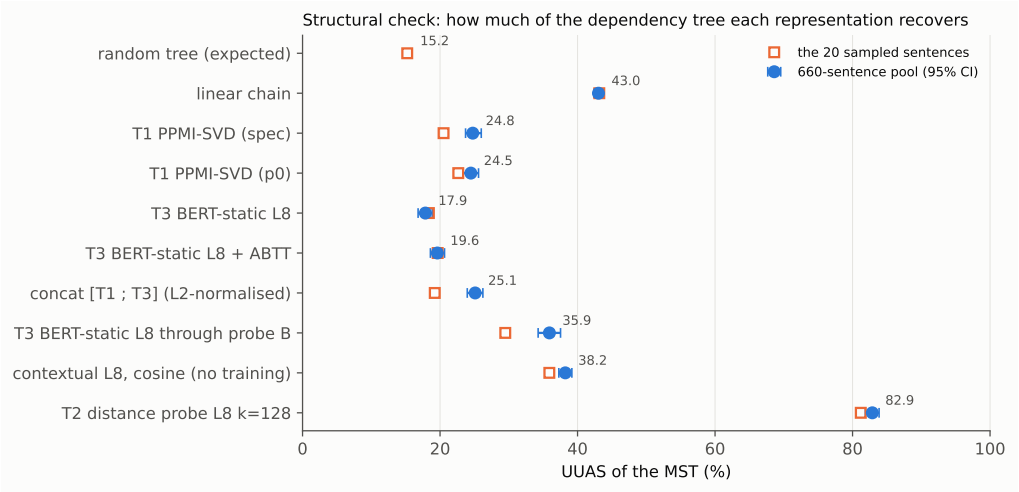


Figure 7: The diagnostic ladder (Table 7): pool UUAS with 95% CI (filled) and the 20 sentences (open squares).

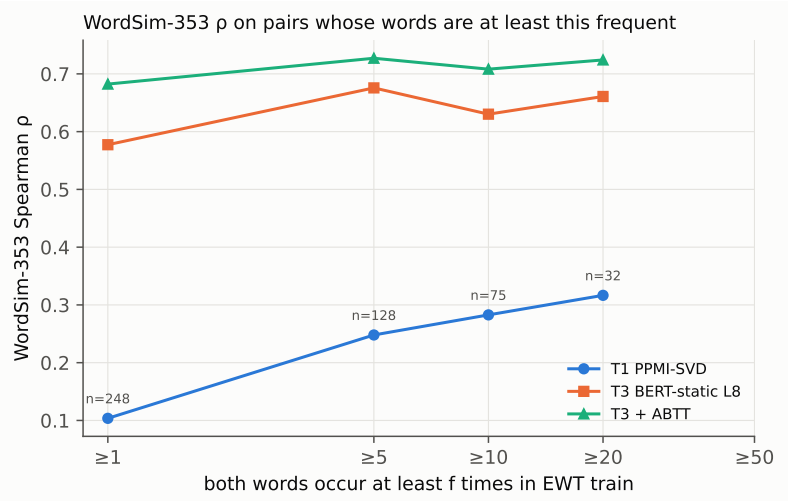
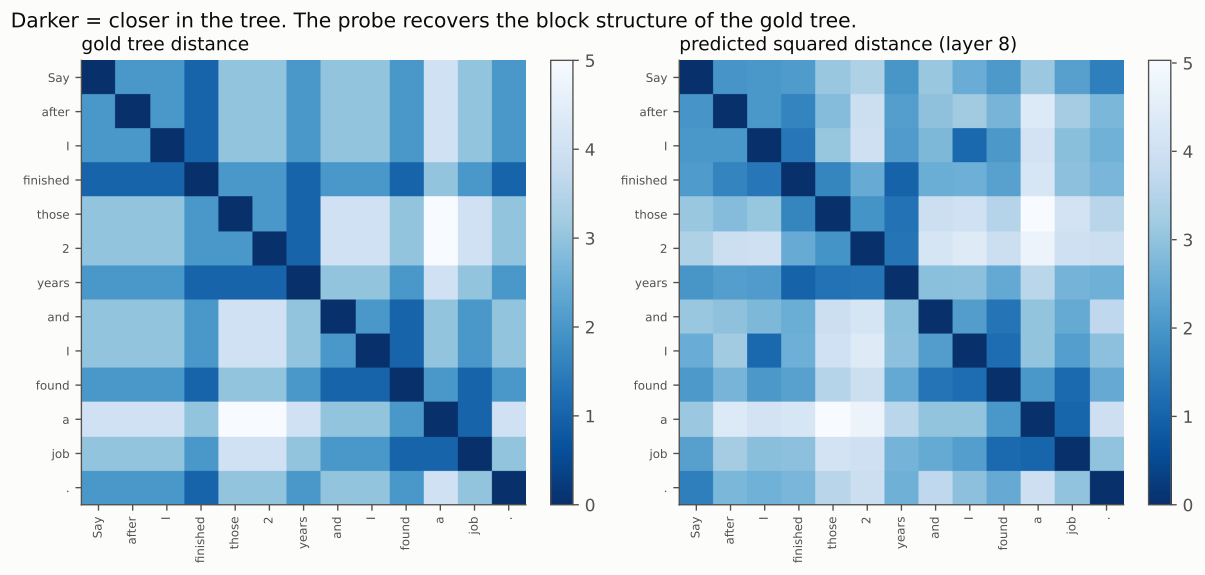


Figure 8: WS-353 ρ on pairs whose two words occur at least f times in EWT train (n = number of pairs).



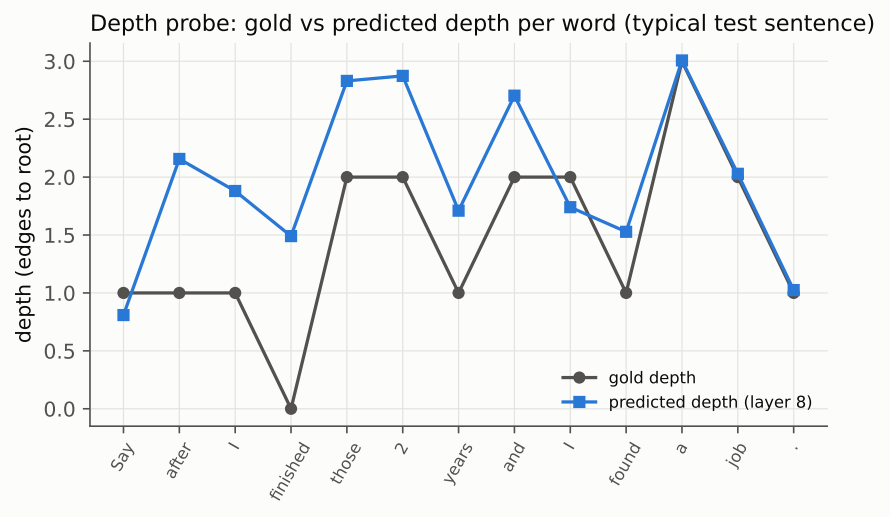


Figure 9: T2 on the median sentence: gold tree distances vs. predicted squared distances (top), gold vs. predicted depth (bottom).